

LectureLens — Brief Project Description

Fully Offline Lecture Copilot Accelerated for Snapdragon X Series Laptops

1. Executive Summary & Pitch

LectureLens is a 100% offline, privacy-first lecture copilot engineered specifically for Qualcomm Snapdragon X Series Windows-on-ARM laptops. It transforms live classroom microphone audio streams or recorded lecture files (WAV/MP3/FLAC) into full transcriptions, executive summaries, essential key term glossaries, interactive multiple-choice quizzes, active recall flashcards, and cross-session semantic vector search — all executing completely on-device without sending a single byte over the internet.

2. Problem Statement & Solution

Over 70% of higher education lectures occur in crowded university halls where campus Wi-Fi deadzones or cellular throttling render cloud-based AI tools completely unusable. Furthermore, students and professors are increasingly hesitant to upload proprietary course content or confidential research audio to third-party cloud APIs.

LectureLens solves this by delivering zero-cloud-dependency on-device AI inference. By harnessing the power of the Qualcomm Hexagon NPU via the Qualcomm QNN Execution Provider inside ONNX Runtime and Foundry Local NPU endpoints, LectureLens processes dense neural speech-to-text and large language model workloads with minimal battery consumption and zero latency.

3. Key Capabilities & Student Features

-  **Live Speech Recognition:** Neural transcription via Whisper STT with sub-second chunking and zero cloud reliance.
-  **Executive Notes & Key Terms:** Automated bullet summaries, glossary definitions, and single-paragraph revision summaries.
-  **Interactive Quizzes:** Schema-validated multiple choice quizzes with instant on-device answer checking and explanations.
-  **Active Recall Flashcards:** Flip-card study modules built for rapid exam revision.
-  **Cross-Session Hybrid Search:** Combines ONNX MiniLM vector embeddings with BM25 keyword matching across all stored lectures.
-  **Strict Offline Guarantee:** Includes socket guard verification enforcing 0 external cloud network requests.

4. Hardware Acceleration & Dual-Engine Architecture

LectureLens features a resilient dual-engine architecture designed for Snapdragon hardware while remaining fully compatible with non-Snapdragon development environments:

- **Speech-to-Text Pipeline (stt_qnn.py):** Wraps ONNX Runtime + Qualcomm QNN Execution Provider (HTP NPU backend) for Whisper base models, with a PyTorch Whisper CPU fallback.
- **LLM Generation Engine (llm_foundry.py):** Connects to local Foundry Local /v1/chat/completions endpoint for NPU Instruct models, with an Ollama local CPU fallback.

5. Verified Performance & Benchmark Metrics

Metric / Component	Runtime / Backend	Measured / Projected Value	Status
STT WER Accuracy	PyTorch Whisper CPU	10% WER (0.10)	Measured (Dev Machine)
STT Speed (RTF)	ONNX QNN / PyTorch	~0.166 RTF CPU ~0.021 RTF NPU	Measured / NPU Projected
LLM Throughput	Ollama phi3:mini / NPU	4.93 tok/sec CPU ~42.5 tok/sec NPU	Measured / NPU Projected
Search Query Latency	ONNX MiniLM CPU	< 5 ms / passage	Measured (Dev Machine)
Offline Network Guard	Python Socket Guard	0 Cloud Network Calls	Verified (Pass)
Test Suite Pass Rate	pytest Framework	24 / 24 Tests Passed	Verified (Pass)

6. Conclusion & Repository Reference

LectureLens bridges the gap between high-performance local AI and practical classroom utility. Source code, verification scripts, pitch deck, and documentation are available at:

GitHub Repository: <https://github.com/suvakovan/SNAP-DRAGON->